

Improving HCC surveillance with abbreviated MRI: A call to integrate and innovate?

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See Article, pages 461–470

DOI of original article: <https://doi.org/10.1016/j.jhep.2024.03.048>



Hepatocellular carcinoma (HCC) remains a formidable challenge in oncology, largely due to its asymptomatic nature in early stages and poor prognosis once advanced. Despite limited evidence, current recommendations for the surveillance of HCC are based on semiannual liver ultrasound (with or without serum alpha fetoprotein) in patients at high risk of liver cancer, mostly those with cirrhosis and subgroups with chronic hepatitis B infection.^{1–3} However, the sensitivity of this strategy is suboptimal for the detection of HCC at the earliest possible stage. The use of ultrasound has been challenged by several meta-analyses^{4,5} or prospective follow-up studies reporting on a poor sensitivity for the detection of very early-stage HCC. Several factors have been proposed to explain this suboptimal performance, including the difficult liver visualization in patients with obesity and/or MASLD, a nodular appearance in a subset of cirrhotic livers, the skills and expertise of the operators, and poor adherence to surveillance programs.

For all these reasons, many argue that other imaging modalities, associated with better detection rates, should be favored and endorsed by HCC guidelines. Based on excellent rates of detection of focal liver lesions with MRI, it is considered the best alternative candidate for surveillance. However, a full liver MRI examination with dynamic multi-phase contrast-enhanced sequences is time consuming and costly, and cannot be realistically and cost-effectively repeated for HCC surveillance. Abbreviated MRI (AMRI) corresponds to the acquisition of a limited number of sequences with a high detection rate. The theoretical benefits of AMRI over conventional MRI are a reduced acquisition time (≤ 10 minutes) with improved time- and cost-effectiveness, while maintaining a better performance, especially a better negative predictive value, than ultrasound. Numerous AMRI protocols may be performed, including T1-weighted, T2-

weighted, and diffusion-weighted infusion sequences, with or without contrast administration.⁶

Multiple published studies have reported promising per-patient results with AMRI, but they should be interpreted with caution. Indeed, most were not performed during surveillance but were simulated retrospectively, considering a subset of sequences extracted from a complete MRI scan performed in small cohorts. Data from prospective actual surveillance cohorts are scarce.^{7–9} Therefore, current evidence is extrapolated from heavily HCC-enriched cohorts that were not representative of surveillance populations. At best, these studies can be seen as proof-of-concept research, but their results cannot be extrapolated to real-life prospective surveillance strategies. In addition, most were published by Asian groups, with at-risk populations that were different from Western populations. Finally, no existing longitudinal studies directly compare the different AMRI acquisition protocols, or, more importantly, AMRI to ultrasound, in particular in relation to the cost and availability of AMRI.

In this issue of the *Journal of Hepatology*, Kim, Yoon and colleagues present the MAGNUS-HCC study, which aims to tackle several of these limitations and concerns.¹⁰ They conducted a prospective multicenter observational study combining biannual ultrasound and annual non-contrast AMRI (NC-AMRI) for the detection of early-stage HCC. The population was selected to have an estimated annual risk of HCC greater than 5%. The final analysis included 208 participants with cirrhosis, mostly caused by chronic HBV infection. After a median follow-up period of 30 months, 34 HCCs were detected in 31 patients (annual incidence rate of 8%). The sensitivity of annual NC-AMRI was not significantly higher than that of biannual ultrasound (71.0% vs. 45.2%, $p = 0.077$). Nevertheless, 12 participants with 13 HCCs were exclusively detected by NC-AMRI, and when both NC-AMRI and ultrasound were utilized, no HCCs

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Received 26 April 2024; accepted 29 April 2024; available online 9 May 2024

<https://doi.org/10.1016/j.jhep.2024.04.035>



Table 1. Ongoing trials testing abbreviated MRI for HCC surveillance.

Trial identification	Country	Patients phenotype	Planned inclusion	Randomization	Comparison	Health economic endpoint	HCC risk stratification	Completion date
NCT03731923	USA	High risk cirrhosis	4,700	Yes	US + AFP	No	Yes	2031
NCT05095714	South Korea	High risk HBV or HCV	257	No (single arm)	US	No	No	2031
	France	High risk cirrhosis (non-viral, viral inactive)	944	Yes	US	Yes	Yes	2028
NCT05657249	South Korea	High risk HBV or HCV	199	No (single arm)	US	No	Yes	2028
NCT06312826	South Korea	High risk cirrhosis	806	Yes	US	No	Yes	2027
NCT05828446	Switzerland	Cirrhosis, high risk HBV	330	No (single arm)	US	No	No	2027
NCT04455932	Australia, New Zealand	Cirrhosis (and poor US visualization)	476	No (single arm)	US	No	No	2027
NCT05429190	Netherlands	Cirrhosis/ACLD	470	No (single arm)	US	Yes	No	2026
NCT04539717	USA	Cirrhosis	820	No (single arm)	US	Yes	No	2025
NCT05716620	India	High risk cirrhosis	320	No (single arm)	US	No	No	2025
NCT04288323	USA	Cirrhosis	150	No (single arm)	US	No	No	2023

Search updated on April 26 2024.

ACLD, advanced chronic liver disease; HCC, hepatocellular carcinoma; US, ultrasound.

were detected solely by ultrasound. However, four participants with four HCCs were identified only by ultrasound, which is surprising given the size of said tumors (mean size of 27.3 ± 21.6 mm [range, 12–59 mm]), larger than those detected by NC-AMRI alone (mean 15.1 ± 8.1 mm [range, 6–36 mm]) or by both ultrasound and NC-AMRI (mean 17.6 ± 4.9 mm [range, 10–28 mm]). Additionally, NC-AMRI achieved a significantly higher diagnostic yield (4.26% vs. 1.43%) without increasing the false referral rate (2.91% vs. 3.06%). Finally, alternating between ultrasound and NC-AMRI every 6 months significantly improved detection rates, suggesting this could be an optimal surveillance strategy for high-risk patients.

This study presents highly anticipated findings that prospectively outline the performance of HCC surveillance based on liver AMRI examination in high-risk patients, mostly with HBV-related cirrhosis. The findings are important for the hepatology and radiology communities, and for patients undergoing surveillance, but must be interpreted with appropriate caution. This is an observational study (not a randomized trial) and therefore the findings can only be regarded as preliminary. The risk-based strategy makes sense in the era of personalized medicine. The caveat is that the results, derived from a high-risk cohort (estimated $\geq 5\%$ annual HCC risk), cannot be extrapolated to the general surveillance population. Indeed, the annual HCC risk in most modern cohorts is as low as 1%, questioning the cost-effectiveness of surveillance with AMRI, even if the results of the study were to be confirmed by ongoing trials (Table 1). Furthermore, the study identified only BCLC 0 or A stage cases, markedly different from European cohorts where approximately one-third of patients are diagnosed at stages beyond BCLC A, which may suggest a shortfall in screening practices. However, the observed range in HCC sizes show that tumors larger than 5 cm were detected. A variety of factors could account for these differences compared to Western populations, such as the exclusion of non-compliant patients, the proficiency of Korean radiologists, and differences in patient body types. Furthermore, as mentioned by the authors, they did not blind radiologists to prior NC-AMRI results due to ethical issues, potentially aiding in the identification of new lesions on ultrasound.

The interpretation of AMRI in the setting of surveillance raises questions. The liver may be heterogeneous or nodular, leading to the detection of non-HCC nodules. A recent study by Nahon and colleagues indicated a higher incidence of non-HCC nodules in cirrhosis compared to HCC.¹¹ While these non-HCC nodules do not impact patients' prognosis, they trigger substantial costs. The introduction of AMRI protocols could increase the detection of HCC precursors (e.g., regenerative or dysplastic nodules), leading to more recall examinations, follow-up and biopsies, false positive results and possible overdiagnoses. In the study by Kim, Yoon and colleagues, the positive predictive values of biannual ultrasound and annual AMRI were 31.8% and 59.5%, respectively, suggestive of a substantial number of false positives. After recall dynamic liver CT or gadoxetic acid-enhanced MRI, the authors reported a false-positive rate of 16.8% comparable to the 19.2% reported by Nahon and colleagues.¹¹

Kim, Yoon and colleagues also discussed the results of a "simulated biannual alternation of US and NC-AMRI". The sensitivity of this strategy was 83.9%, significant higher compared to biannual ultrasound alone, without a significant difference in the false referral rate. Such alternative strategies, including less frequent screening with AMRI or alternating AMRI with ultrasound may be considered to "get the best from both worlds". There is increasing recognition that the value of cancer screening programs must also consider the physical, financial, and psychological harm of these tests, and the extent of AMRI screening-related harms are currently unknown.

In conclusion, while this study marks a significant step forward in the quest for optimal HCC surveillance, it is merely the beginning of what must be a sustained, collaborative effort to refine and implement strategies that will ultimately reduce the burden of this deadly disease.

Liver AMRI will undoubtedly be included in future surveillance strategies, but a thorough and accurate evaluation of its accessibility, acceptability and cost-benefit ratio is needed before considering its widespread implementation in clinical practice. The ongoing trial will further provide valuable and much needed evidence.

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Financial support

The author did not receive any financial support to produce this manuscript.

Conflict of interest

The author of this study declares that they do not have any conflict of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.04.035>.

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